Sports Physical Therapy Database Project

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CSCI 411 Database Theory and Design

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1. Introduction

The purpose of this project is to design and implement a relational database system for a sports physical therapy clinic using Microsoft Access. The database will help manage patient information, therapist information, therapy types, and scheduled therapy sessions.

The system will allow the clinic to track patient therapy assignments, therapist qualifications, session attendance, and patient progress.

2. Database Design

The database consists of six tables: Patient, Therapist, TherapyType, PatientTherapy, TherapistTherapy, and TherapySession.

The tables were designed to organize information and reduce unnecessary data duplication. Primary keys uniquely identify records, while foreign keys connect related tables and maintain referential integrity.

2.1 Entity Relationship Diagram (ERD)

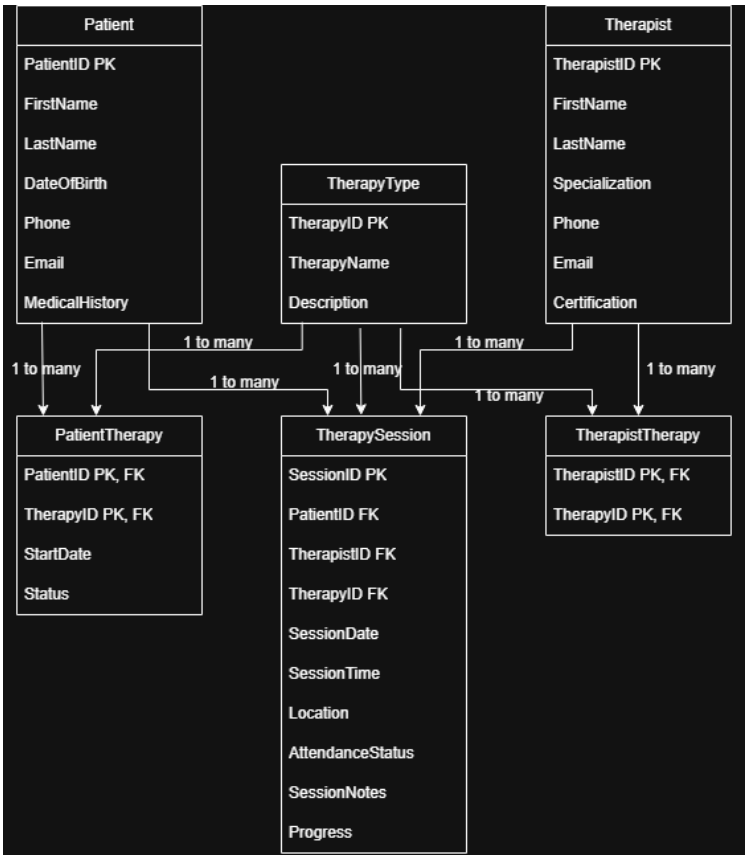


Figure 1. Sports Physical Therapy ER Diagram

The ER diagram shows the six tables, their attributes, primary keys, foreign keys, and relationships. The database contains seven one-to-many relationships. The PatientTherapy and TherapistTherapy junction tables resolve the many-to-many relationships between patients and therapy types, and between therapists and therapy types.

2.2 Table Descriptions

1. Patient: Stores patient information, including names, date of birth, contact information, and medical history.
2. Therapist: Stores therapist information, including names, specializations, certifications, and contact information.
3. TherapyType: Stores available therapy types and their descriptions.
4. PatientTherapy: Connects patients to their assigned therapy types and tracks the start date and status of each assignment.
5. TherapistTherapy: Connects therapists to the therapy types they are qualified to provide.

6. TherapySession: Stores scheduled therapy sessions, including the patient, therapist, therapy type, date, time, location, attendance status, notes, and progress.

3. Microsoft Access Implementation

The sports physical therapy database was implemented using Microsoft Access. Six tables were created to organize patient information, therapist information, therapy types, patient therapy assignments, therapist therapy assignments, and therapy sessions.

Each table uses an appropriate data type for its attributes. AutoNumber fields are used for the primary keys of Patient, Therapist, TherapyType, and TherapySession. Foreign keys use the Number data type with Long Integer field size to connect related records. Composite primary keys are used in the PatientTherapy and TherapistTherapy junction tables to prevent duplicate patient-therapy and therapist-therapy assignments.

Referential integrity was enabled for all relationships to help maintain accurate connections between records and prevent invalid foreign key references.

The database also includes sample records for patients, therapists, therapy types, therapy assignments, and scheduled therapy sessions.

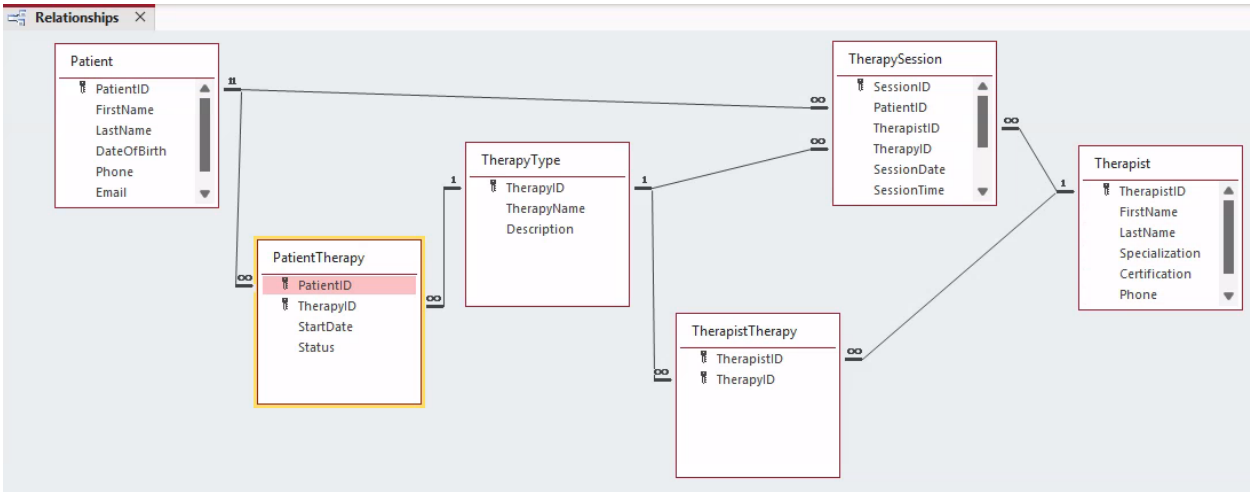


Figure 2. Relationships between the six database tables in Microsoft Access.

4. Forms

Two forms were created in Microsoft Access to make entering and managing information easier.

4.1 Patient Intake Form

The Patient Intake form (frmPatientIntake) allows users to enter patient information, including first name, last name, date of birth, phone number, email address, and medical history. This form provides a more organized way to add patient records without directly editing the table.

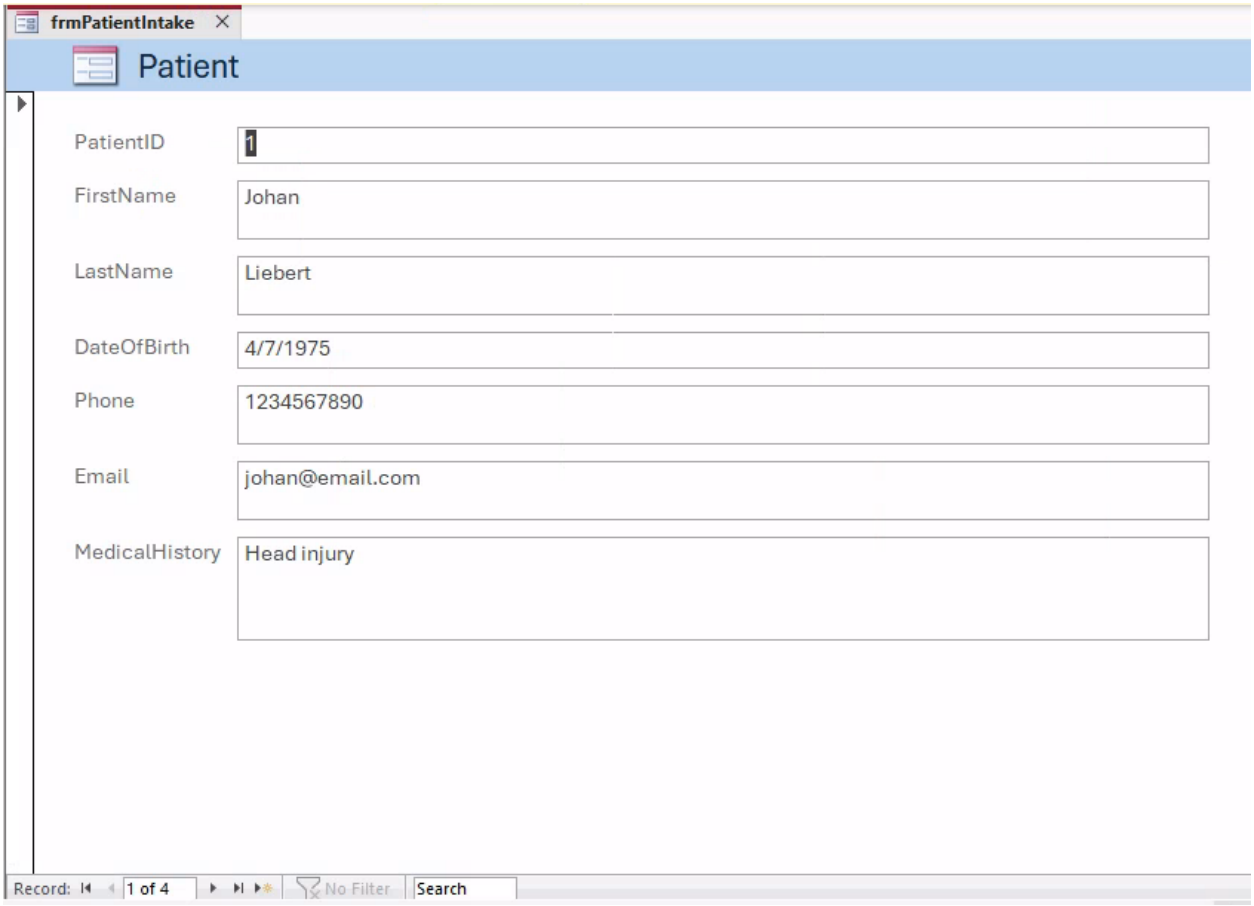


Figure 3. Page 1 of Patient Intake form used to enter patient information.

4.2 Session Scheduling Form

The Session Scheduling form (frmSessionScheduling) allows users to enter and manage therapy session information. The form includes the patient ID, therapist ID, therapy type ID, session date, session time, location, attendance status, session notes, and progress.

This form helps organize scheduled sessions and record information about completed therapy appointments.

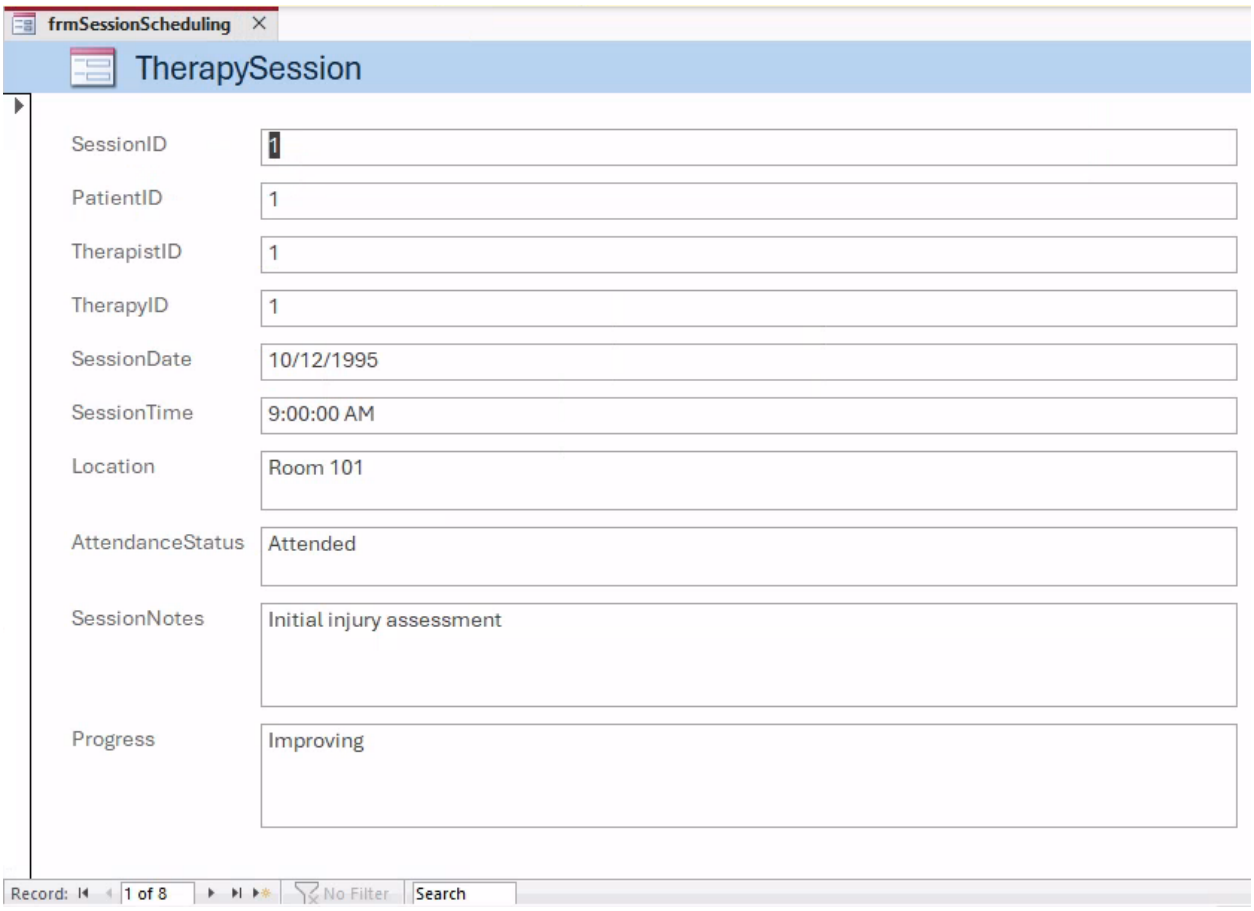


Figure 4. Page 1 of Session Scheduling form used to manage therapy sessions.

5. Queries

Three queries were created to retrieve and summarize information from the database.

5.1 Patient Progress Query

The Patient Progress query (qryPatientProgress) combines information from the Patient, TherapySession, Therapist, and TherapyType tables. It displays the patient's name, therapy type, session date, assigned therapist, attendance status, session notes, and progress.

This query allows users to review a patient's therapy history and monitor their progress across different sessions.

qryPatientProgress

Patient.First	Patient.Last	TherapyName	SessionDate	Therapist.Fi	Therapist.La	Attendance	SessionNotes	Progress
Johan	Liebert	Injury Rehabilitation	10/12/1995	Kenzo	Tenma	Attended	Initial injury assessment	Improving
Johan	Liebert	Strength Training	10/14/1995	Tony	Chopper	Attended	Lower body strength exercises	Good progress
Nina	Fortner	Mobility Exercises	10/13/1995	Tsunade	Hokage	Attended	Shoulder mobility exercises	Improving
Nina	Fortner	Balance Training	10/15/1995	Tony	Chopper	Scheduled	Balance assessment	Not yet evalutated
Heinrich	Lunge	Injury Rehabilitation	10/16/1995	Kenzo	Tenma	Attended	Knee rehabilitation exercises	Moderate progress
Heinrich	Lunge	Balance Training	10/19/1995	Tsunade	Hokage	Scheduled	Balance training session	Not yet evaluated
Wolfgang	Grimmer	Mobility Exercises	10/20/1995	Tsunade	Hokage	Scheduled	Ankle mobility exercises	Not yet evaluated
Wolfgang	Grimmer	Strength Training	10/21/1995	Tony	Chopper	Scheduled	Strength training session	Not yet evaluated

Figure 5. Patient Progress query showing therapy session information.

5.2 Therapist Workload Query

The Therapist Workload query (qryTherapistWorkload) combines information from the Therapist and TherapySession tables. It groups sessions by therapist and counts the number of sessions assigned to each therapist.

This query helps determine how many therapy sessions each therapist is responsible for.

qryTherapistWorkload

FirstName	LastName	CountOfSes
Kenzo	Tenma	2
Tony	Chopper	3
Tsunade	Hokage	3

Figure 6. Therapist Workload query showing the number of sessions assigned to each therapist.

5.3 Weekly Session Summary Query

The Weekly Session Summary query (qryWeeklySessionSummary) combines information from the Patient, Therapist, TherapyType, and TherapySession tables. It retrieves session IDs, patient names, therapist names, therapy types, session dates, session times, locations, and attendance statuses.

This query serves as the data source for the Weekly Session Summary report.

SessionID	Patient.First	Patient.Last	Therapist.Fi	Therapist.La	TherapyNan	SessionDate	SessionTime	Location	Attendance
1	Johan	Liebert	Kenzo	Tenma	Injury Rehabili	10/12/1995	9:00:00 AM	Room 101	Attended
2	Johan	Liebert	Tony	Chopper	Strength Traini	10/14/1995	10:00:00 AM	Gym A	Attended
3	Nina	Fortner	Tsunade	Hokage	Mobility Exerci	10/13/1995	11:00:00 AM	Room 102	Attended
4	Nina	Fortner	Tony	Chopper	Balance Trainir	10/15/1995	1:00:00 PM	Gym B	Scheduled
5	Heinrich	Lunge	Kenzo	Tenma	Injury Rehabili	10/16/1995	9:30:00 AM	Room 101	Attended
6	Heinrich	Lunge	Tsunade	Hokage	Balance Trainir	10/19/1995	2:00:00 PM	Gym B	Scheduled
7	Wolfgang	Grimmer	Tsunade	Hokage	Mobility Exerci	10/20/1995	10:30:00 AM	Room 102	Scheduled
8	Wolfgang	Grimmer	Tony	Chopper	Strength Traini	10/21/1995	11:30:00 AM	Gym A	Scheduled

Figure 7. Weekly Session Summary report displaying scheduled therapy sessions, patient and therapist information, and attendance status.

6. Reports

A report named rptWeeklySessionSummaryFinal was created using the Weekly Session Summary query. The report lists therapy sessions in ascending order by session date and displays the patient, assigned therapist, therapy type, scheduled date and time, location, and attendance status.

The report provides a convenient overview of scheduled and completed therapy sessions. It can be used to review clinic appointments and see which sessions have been attended or are still scheduled.


rptWeeklySessionSummaryFinal

rptWeeklySessionSummaryFinal

Patient.FirstName	Patient.LastName	SessionDate	onID	Therapist.FirstNam	Therapist.LastNa	TherapyName	SessionTime	Location	AttendanceSta
Heinrich	Lunge								
		10/16/1995	5	Kenzo	Tenma	Injury Rehabilitation	9:30:00 AM	Room 101	Attended
		10/19/1995	6	Tsunade	Hokage	Balance Training	2:00:00 PM	Gym B	Scheduled
Johan	Liebert								
		10/12/1995	1	Kenzo	Tenma	Injury Rehabilitation	9:00:00 AM	Room 101	Attended
		10/14/1995	2	Tony	Chopper	Strength Training	10:00:00 AM	Gym A	Attended
Nina	Fortner								
		10/13/1995	3	Tsunade	Hokage	Mobility Exercises	11:00:00 AM	Room 102	Attended
		10/15/1995	4	Tony	Chopper	Balance Training	1:00:00 PM	Gym B	Scheduled
Wolfgang	Grimmer								
		10/20/1995	7	Tsunade	Hokage	Mobility Exercises	10:30:00 AM	Room 102	Scheduled
		10/21/1995	8	Tony	Chopper	Strength Training	11:30:00 AM	Gym A	Scheduled

Thursday, November 5, 2026

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Figure 8. Weekly Session Summary report displaying therapy session information.

7. Conclusion

The sports physical therapy database was created to help organize patient, therapist, therapy, and session information in one place. Using Microsoft Access, six related tables were built along with relationships, forms, queries, and reports.

The forms make it easier to enter patient and therapy session information. The queries help users look up patient progress and keep track of therapist workloads. The weekly session report also provides a clear overview of scheduled and completed appointments.

Overall, the database gives a structured way to manage information for a sports physical therapy clinic, and it also shows how a relational database can be used to organize and retrieve information efficiently.